

PROJECT DESIGN

4.1 Problem Solution Fit

Project Title

Optimising User, Group and Role Management with Access Control and Workflows

Introduction

Many organizations face challenges in managing users, groups, and roles efficiently. Manual access management often leads to permission errors, delayed approvals, and security risks. This project provides an automated solution using ServiceNow to simplify user administration, enforce secure access control, and streamline approval workflows.

Problems Identified

Manual user and group administration.

Delays in granting or removing access.

Incorrect role assignments.

Unauthorized access to confidential data.

Repetitive approval processes.

Lack of proper audit tracking.

Increased administrative workload.

Proposed Solution

The proposed ServiceNow application includes:

Automated user provisioning and deprovisioning.

Group-based access management.

Role-Based Access Control (RBAC).

Access Control Lists (ACLs) for secure data access.

Workflow automation using Flow Designer.

Email notifications for approvals.

Audit logs and reporting for compliance.

Problem–Solution:

The proposed solution automates user, group, and role management in ServiceNow by implementing Role-Based Access Control (RBAC), Access Control Lists (ACLs), and workflow automation. It centralizes user administration, ensures accurate role assignments, secures access to sensitive data, and automates approval processes using Flow Designer. This reduces manual effort, minimizes errors, improves security, and enhances overall operational efficiency and compliance.

Expected Benefits

- Enhanced system security.
- Faster access approval process.
- Reduced manual effort.
- Improved compliance and audit readiness.
- Better productivity and user experience.
- Accurate role and permission management.

Conclusion

This project improves the efficiency of user, group, and role management by automating access control and approval workflows in ServiceNow. It enhances security, reduces manual errors, and ensures better governance through centralized access management.

Problem-Solution Fit Canvas — Optimising User, Group & Role Management with Access Control & Workflows

Define CS, fit into CC, explore AS	1. Customer Segment(s) CS IT Administrators, Employees/Requesters, Approvers/Managers & Compliance teams in mid-large enterprises using ITSM/IAM platforms.	2. Customer Constraints CC Limited IT staff bandwidth, strict compliance/audit requirements, licensing budget limits, resistance to changing manual processes.	3. Available Solutions AS Manual spreadsheets for role tracking, native access request forms, generic IAM tools, email-based approval chains.
	4. Jobs-to-be-Done, Pains & Gains JPG Job: grant/revoke access quickly & accurately. Pains: delays, over-provisioning, audit failures. Gains: faster onboarding, less risk.	5. Root Cause RC Lack of centralized, rule-based role/group management and automated workflows leads to manual, error-prone provisioning.	6. Existing Behaviour EB Admins manually assign roles via spreadsheets/tickets; approvals happen over email; periodic manual audits to find excess access.
Focus on JPG, tap into RC, understand EB	7. Triggers TR New hire onboarding, role change/promotion, project assignment, audit finding excess access, employee offboarding.	8. Emotions EL Frustration with delays, anxiety about compliance risk, relief when access works correctly, stress during audits.	9. Channels of Behaviour CB Service desk portal, email requests, IT admin console, HR system triggers, compliance dashboards.
	10. Adoption Barriers: AB Change management resistance, integration effort with existing HR/ITSM systems, training needs for the new workflow tool, initial setup & configuration time.		

4.2 Proposed solution:

S.No	Parameter	Description
1.	Problem Statement (Problem to be Solved)	Organizations face challenges in managing users, groups, and roles efficiently. Manual access management leads to security risks, duplicate records, permission errors, and delays in user provisioning and approval processes.
2.	Idea / Solution Description	The proposed solution is a centralized User, Group, and Role Management system in ServiceNow that automates user provisioning, role assignment, access control, and approval workflows to improve security and operational efficiency.
3.	Novelty / Uniqueness	The solution provides automated role assignment, workflow-based approvals, centralized access control, and efficient user lifecycle management, reducing manual effort and improving accuracy.

4.	Social Impact / Customer Satisfaction	The system enhances security, transparency, and user satisfaction by ensuring timely access, reducing errors, and providing faster approval and access management processes.
5.	Business Model (Revenue Model)	The solution is implemented using the ServiceNow platform and follows a Software-as-a-Service (SaaS) model, enabling scalable and cost-effective enterprise access management.
6.	Scalability of the Solution	The solution is highly scalable and can efficiently manage a growing number of users, groups, roles, and workflows while maintaining performance, security, and compliance.

4.3 Solution Architecture

The solution architecture consists of Users, Groups, Roles, Access Control Rules (ACLs), Workflows, and the ServiceNow Platform. Users submit access requests, workflows route them for approval, roles are assigned based on permissions, groups organize users, ACLs enforce security, and all activities are logged for monitoring and auditing.

